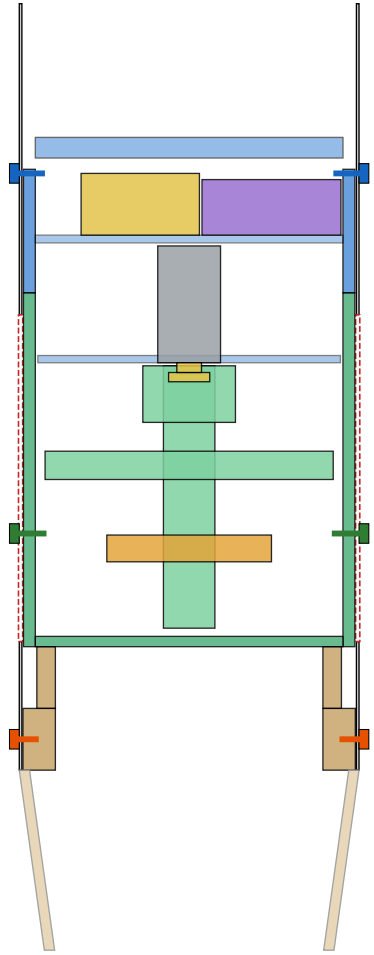
V5 complete concept: the modular airbrake

The whole idea on five pages (concept drawings). Status: BUILT in CAD as mode 5 and checked, not printed. What is built: V5_assembly_guide.pdf.

What V5 is, in one paragraph

The airbrake will break every flight, so V5 is built to swap the part that breaks. One BT80 tube, cut once, is the skin. Two inserts slide into it: the MODULE (housing, rotor, four fins: the part that breaks) and the HEAD (SG90 servo, Mercury V1, regulator, 2S battery, switch: reusable). The head sits above the module and stays in the tube; the servo horn drops into the rotor hub slot, so no wire crosses a joint. Below sits the TRANSITION (BT80 to BT70). Everything is held by plain M3 bolts and nuts, radial, from the outside through the tube, on the BT80 side only. The BT70 side is not touched.

Concept, to scale



Your hard rules (they decide the design)

Rule	What it means here
M3 bolts and nuts wherever possible	no heat-set inserts, no self-tapping into plastic
All bolts radial, from the outside	through the BT80 tube; BT80 side only; nothing from below
Never touch the BT70 side	the transition below the BT80 plug and shoulder is yours; cone and sleeve here are placeholders
No snap hooks, no bands	nothing that flexes or wraps around to hold a part
Housing in one piece	no seam, no seam screws: the rotor drops in from the top

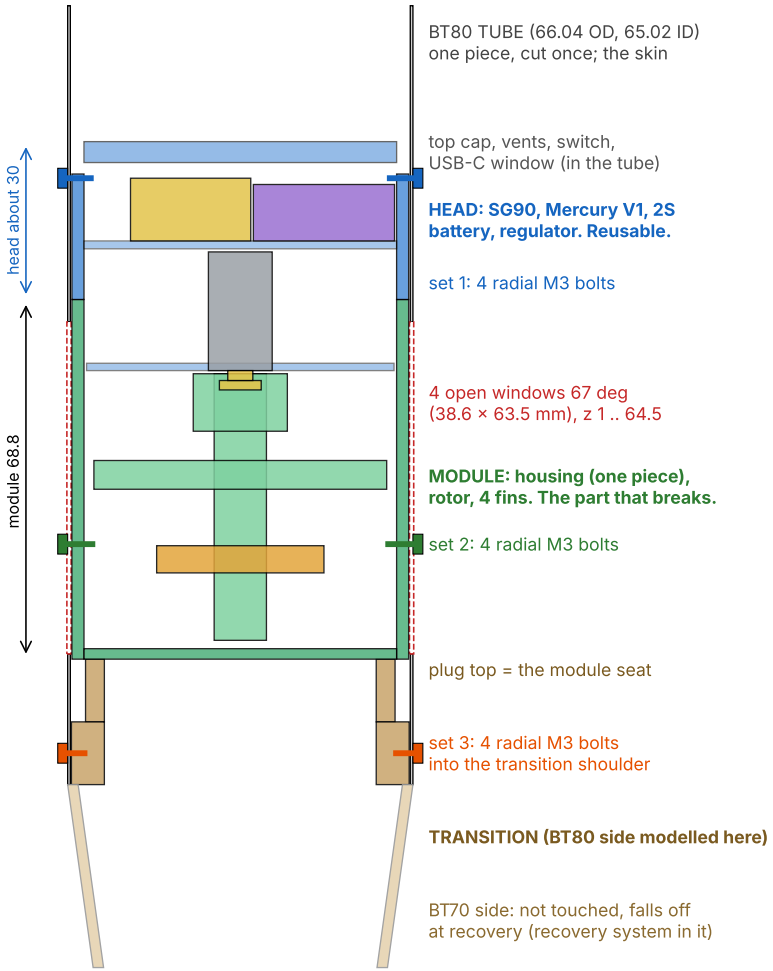
What changes from V4 to V5

	V4 (built)	V5 (planned)
Where the skin is	BT80 pieces glued to the fins	one BT80 tube is the skin; windows left open (tape optional)
Swap	cut and rebuild the whole airbrake	slide out the module; 8 bolts; the head is never touched
Housing	two halves, 4 seam screws, pegs	one piece; limiter moved above the bearing ring
Electronics	bay and sled inside the housing, wires cross the housing	in the head: no wire crosses a joint; V4 vents and cable hole disappear
Fasteners	27 screws (4 seam, 4 cap, 4 hinge M4, 4 boss, 8 tube support, 3 M2)	about 15: 12 radial M3 + nut, hinges M3 + nut (counted in the 12), 3 M2 servo
Bottom	bay, lower sleeve, bonded ring	transition plug and shoulder (BT80 side), bolted
Size and weight	105.4 mm overall, 144 g (estimate)	about 107 mm overall; module about 105 g, head about 55 g (estimates)

The stack and the three bolt sets

Source of truth for the numbers: HANDOFF.md section 0 (concept values; the CAD may move them).

The stack, nose up (full section, concept; z = 0 at the module bottom)



The three independent bolt sets (bolt path B, no sleeve)

Set	What it holds	Where	Nut	Azimuths
1 head	head and lid to tube	tube + head cup wall + lid boss	hex pocket in the lid boss	45, 135, 250, 290
2 module	module to tube	tube + module wall (nut boss R27.6..29.9)	slot open inward	45, 135, 225, 315
3 transition	transition to tube	tube + transition shoulder (R26.0..32.3)	slot in the shoulder	45, 135, 225, 315

All 12 holes are plain M3 clearance holes in the tube (final size set in CAD), cut once from the 1:1 template. A hole is only a hole: a new module reuses the same 12 holes, so holes are reused, never multiplied. What wears is the bolts and nuts (cheap). Bolt: M3×8 button head, 2.0 mm hex key, nut slot 5.7 wide x 2.6 deep.

The module seat is the plug top (R26.0..29.6, 12 mm). The module does not wrap the plug, so the module load goes into the four module-wall bolts only. Estimated shock: 100 g on a 105 g module is about 100 N, 25 N per bolt; an M3 bolt in a 2.7 mm printed wall bears about 100 to 240 N (screening values, not tested).

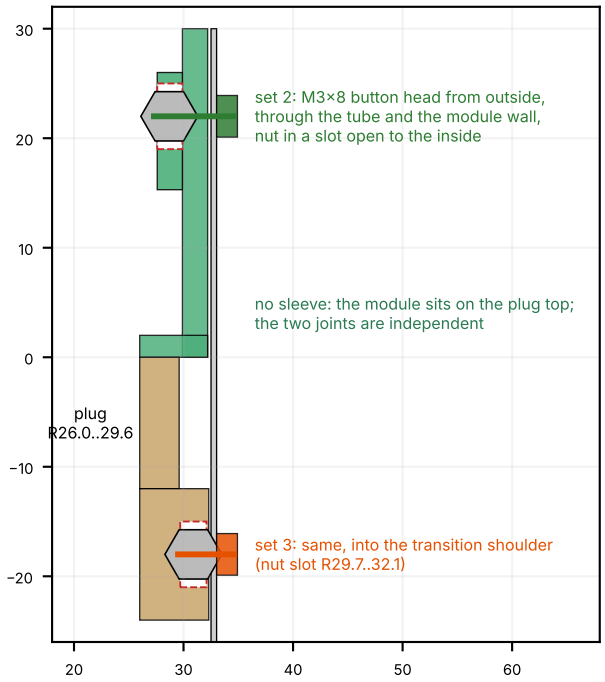
Parts and what is new

Part	Work	Based on
Tube (reference)	4 windows 67 deg x 63.5 mm, 12 clearance holes, switch, USB-C, 3 vents	V4 upper tube
Module housing	one piece; no bay, bosses, supports, ledges, seam; 4 nut bosses; limiter above the ring	V4 housing
Rotor	integral catch caps (no cap screws), hub slot with a lead-in	V4 rotor
Fins	same shape; M3 hinge bolt + nut slots instead of M4	V4 fins
Head	sleeve with 4 bosses and nut slots, cup with servo ledges, plate, single-layer sled, top cap	new (V4 sled, ledges)
Transition (BT80 side)	plug, shoulder, 4 nut slots; cone and BT70 sleeve are placeholders	new
Assembly	mates: module slides along the tube, fins open, horn finds the slot	asm.py

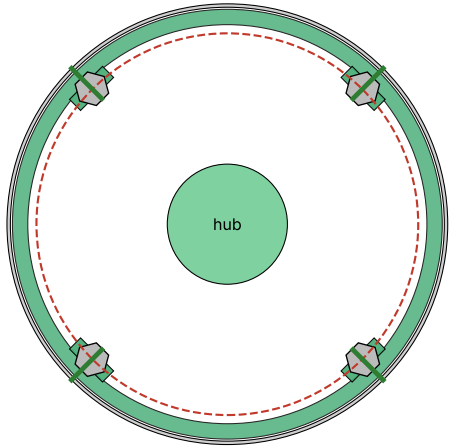
How it fits: the joints, the holes and the tube

Detail sections (concept). Every bolt is radial, from the outside, through the BT80 tube.

A Axial section through two bolts (module and transition)

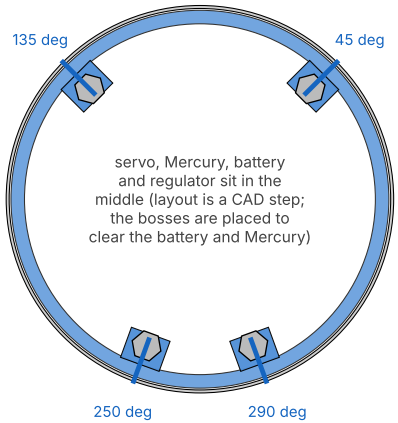


B Plan section at the module bolts (z 22)

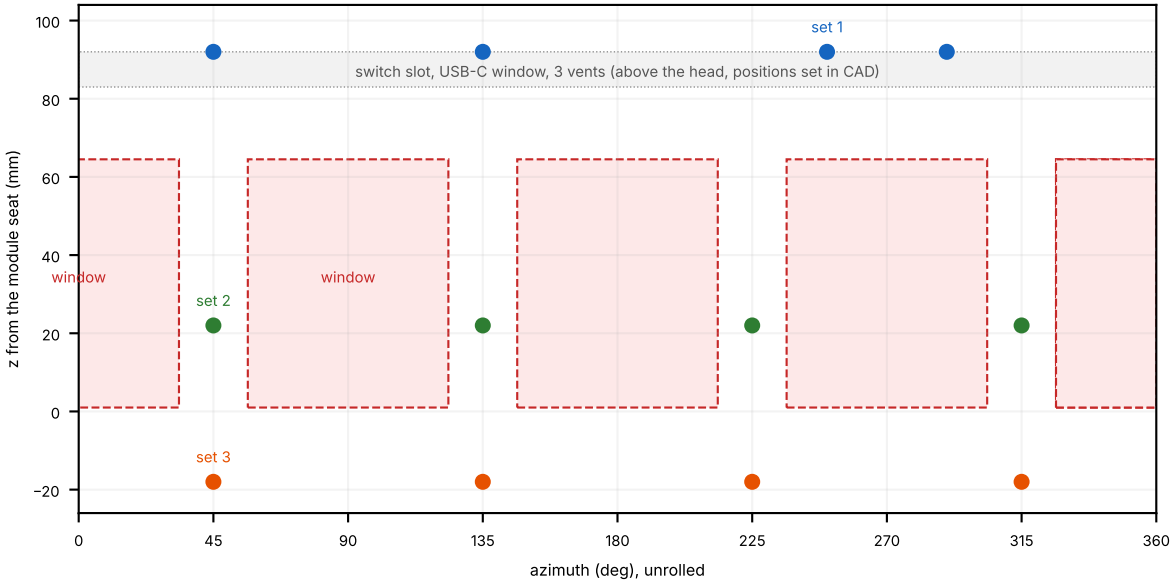


dashed: cam peak R28.6 (module wall inside R29.9; bosses R27.6)
the cam lobes pass the boss height only while the rotor drops in:
lobe azimuth vs boss azimuth (45 + 90k) is a CAD check

C Plan section at the head bolts (set 1, z 74.8)



D The tube unrolled: 4 windows and the 12 clearance holes, cut once



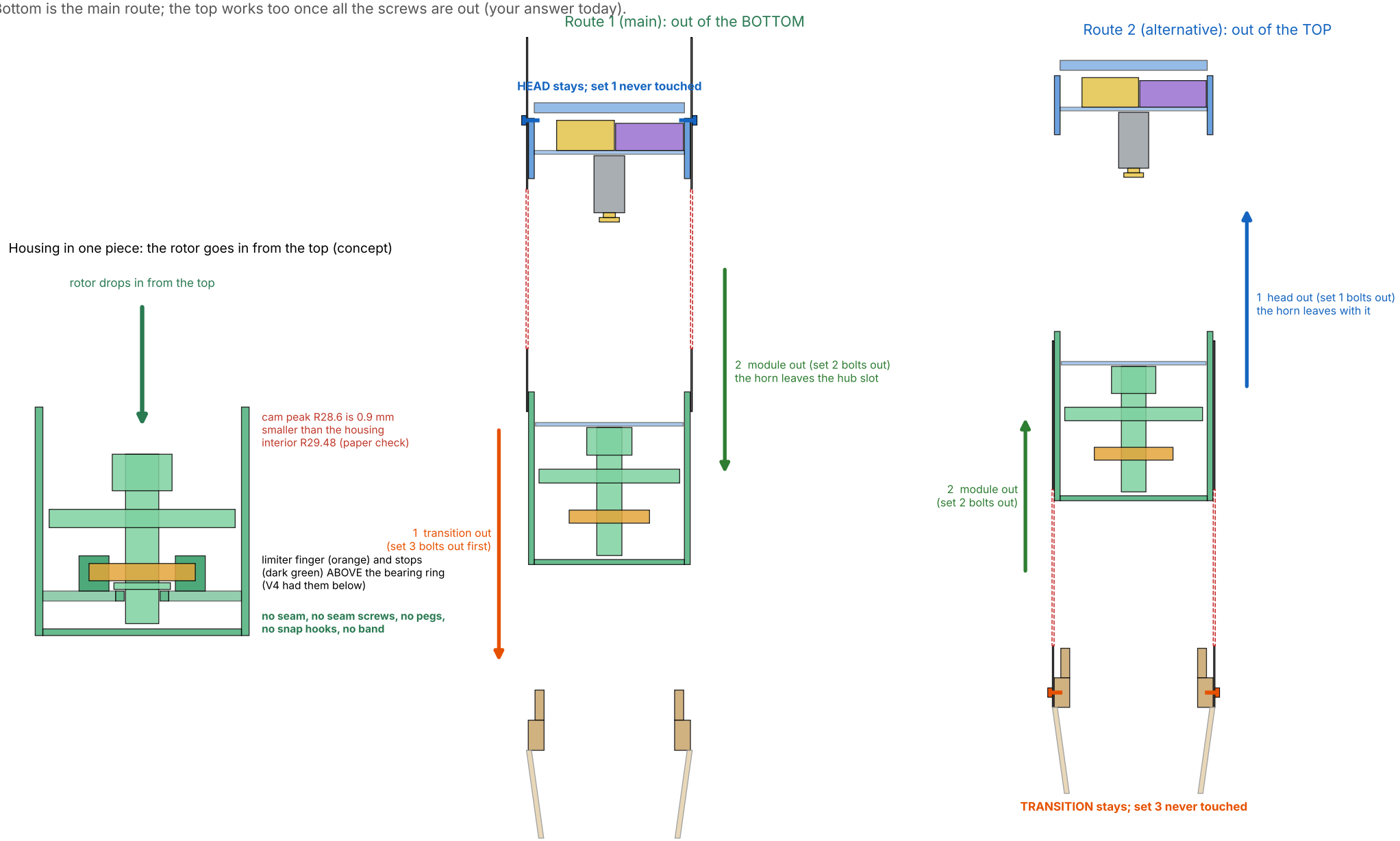
Why nothing is multiplied

Nothing is screwed into plastic. Every hole is a plain M3 clearance hole and every bolt ends in a nut.
The tube has 12 holes: 4 head, 4 module, 4 transition. They are cut once.
The transition has 4 holes and 4 nut slots, the same 4 every time.
Each new module is printed with its own 4 nut slots at the same positions,
so it lines up with the existing tube holes.
Unbolting leaves nothing damaged. The transition is not used up.

Tube features: 4 windows (open; one wrap of foil or Kapton tape is optional,
about 5.6 N suction per window at 80 m/s by the torque-check assumption),
a switch slot, a USB-C window and 3 vents above the head.
Fins stay inside R32.2: a fin with its own skin would not slide into the tube.

One-piece housing and the two ways out

Bottom is the main route; the top works too once all the screws are out (your answer today).



Route 1 (main; the BT70 section is already off): 1 pull the 4 transition bolts and slide the transition out of the bottom. 2 pull the 4 module bolts and slide the module out of the bottom (the horn leaves the hub slot). 3 new module in, nuts in the slots first, 4 bolts. 4 transition back, 4 bolts. 8 bolts; the head is never touched.

Route 2 (alternative): pull the 4 head bolts and lift the head out; pull the 4 module bolts and lift the module out of the top. 8 bolts; the transition stays on. It needs the upper airframe separated from the top of this tube (assumption, please confirm). Both routes work because nothing on the module or the head is wider than the bore or acts as a flange.

Recent decisions, what they replaced, and what is open

Order follows the repository history; the user's own words are in quotes where the handoff recorded them.

#	Decision	Why	Status
1	Make the airbrake modular: swap the whole module	"it will break every flight"	decided
2	One BT80 tube cut once is the skin; no BT80 glued to the fins; windows open (tape optional)	a fin with its own skin cannot slide into the tube	decided
3	Window size is 67 deg x 63.5 mm (38.6 × 63.5), not 43 mm tall	measured on the V4 reference tube; earlier drafts were wrong	corrected
4	No lower BT80 tube; the "transition ring" is the top of the transition; BT70 OD 56.3	the BT70 side holds the recovery system and falls off	corrected
5	M3 bolts + nuts, no heat-set inserts, no self-tapping	rule	decided
6	All bolts radial, from outside, BT80 side only; screws from below rejected	rule: "I do not have much space in the transition"	decided
7	Bolt path B: three independent sets of 4, no module sleeve over the plug	sleeve adds 12 mm, clamps the part that breaks	decided
8	Housing in one piece ("ok ye try the housing one piece"); no snap hooks, no band	removes seam, 4 seam screws, pegs	verified in CAD: rotor drops in at any angle 0..60 deg
9	Head is a separate reusable insert; no wire crosses a joint	V4 vents and cable hole disappear	decided
10	Module comes out of the bottom mainly; also out of the top when all screws are out	your answer today	decided; design rule added
11	Go-ahead for the CAD build	you asked for this document first, then "go ahead and build V5"	given; V5 built, not printed

Replaced along the way

heat-set inserts with 4 head + 2 index screws (V5_final_plan)
module skin and bayonet options (choice 1 and 2 sheets)
screws from below (transition sketch)
bolt path A with a module sleeve (sketch 2; its screw table and nut-in-slot detail still hold)
"open the top only" as the single swap route (insert concept; now bottom is the main route)

Still open (I will not guess these)

1 (done) V5 CAD built: FeatureScript mode 5, transition BT80 side, head, assembly, checks, exports. Print and fit tests are next.
2 Transition diameters unknown: I model the BT80 plug and shoulder; cone and BT70 sleeve are placeholders; the BT70 ID is needed later.
3 Windows open or taped (tape is optional).
4 Route 2 needs the upper airframe to open at the top of this tube: please confirm this matches your recovery layout.
5 Carried over from V4: battery size, Mercury servo header pins, SG90 horn dimensions, ground-impact FEA, a real print test.

Checks before I call V5 done

V1 to V4 stay volume-identical (regression rebuild)
no printed overlaps, closed or open; the module slides along the tube
fins open through the windows; hinge bolts reachable through them
head goes on and the horn finds the slot (about +-8 deg); both exit routes clear
rotor sweep, latch and cam timing; one-piece drop-in clearances
nut slots clear of the cam during the rotor drop-in
hex key (2.0 mm) reaches all 12 bolts
STL (print orientations: rotor hub down, fins flat, housing upright) and STEP

Fallback if the one-piece housing fails: two halves joined by a sliding dovetail along the seam (no flexing parts, no snap hooks).
Design rules added by today's answer: nothing on the module or the head wider than the bore, no flange or lip that blocks either direction, hub slot open toward the head.